Natural Misère Quotients: Exact Octal Traces, Heap Normal Forms, and Grundy's Game

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Abstract

Siegel's valid-transition-table theorem characterizes abstract finite misère quotients. Natural realization asks for more: one uniform heap rule must generate the table at every heap size. We formulate that requirement as a canonical quotient-valued heap trace. In every reduced valid table the complete option-value set determines its value, and no transition value can occur among its own options. These facts give an if-and-only-if trace criterion, absolute digit obstructions, automatic periodicity for finite no-split quotients, and finite certificates for asserted periodic split traces. Any exact word periodic from N with period p has a realizing code of length at most $3N + 2p$, giving finite synthesis with bounded periodic complexity. The split recurrence is exactly a system of unary Boolean language equations. For arbitrary partial decoders its totality problem is Π^0_1-complete already for 0.7; the reduction deliberately falls outside the valid-table class. For that same code every nontrivial valid table has the universal parity trace. More generally, every two-element split quotient has a period-two heap trace, giving a complete finite mask classification for this target.

On the realization side, every nontrivial finite valid quotient has a one-species heap model with finitely many source-local rules and the uniform tail $H_n \to H_{n-1}$, but an inert-pad theorem prevents a verbatim finite-octal encoding when the quotient has more than two elements. For ordinary subtraction games we give a self-contained derivation from Ferguson's theorem of the complete realization classification: the trivial quotient, the two-element quotient, and the tame quotients $\mathcal{T}_n$. The code with 2^{n-1} consecutive digits 3 realizes $\mathcal{T}_n$ for every $n \geq 2$. For misère Grundy's game we prove exact all-multiplicity quotients of orders 6, 12, and 24 through heaps 12, 13, and 18. A finite full quotient forces a heap with its own squared value among its option values, and hence a nonconstant positive-power cycle. The canonical mate of H_{13} has no representative among Grundy positions. The wild octal-realization family and the full Grundy quotient remain open.

1 Introduction

An abstract finite impartial game can be assembled to realize a prescribed transition table. A finite octal code cannot be assembled heap by heap: its kth digit imposes the same translation-invariant move on every sufficiently large heap. The natural realization problem is to identify the extra algebraic condition imposed by that uniform tail.

The argument has one spine. Transition-table determinism makes the heap word canonical; the octal rule turns that word into a totality problem; exact periodic certificates and normal forms locate the boundary of the problem; and the subtraction classification and Grundy calculations solve the recurrence in two complementary regimes. The main results are grouped accordingly.

Theorem A. *1. A reduced valid transition table assigns at most one value to each option-value set. A finite octal code realizes a finite reduced bipartite monoid exactly when its induced heap trace is total, lies in such a table, and generates the monoid. No transition value is one of its own option values, yielding explicit whole-, one-, and two-remainder exclusions.*

2. *Finite no-split quotients have ultimately periodic heap traces, so exact realization by a fixed no-split code is decidable. An asserted periodic split trace also has a finite exact certificate. Every exact word periodic from N with period p has a realizing code of length at most $3N + 2p$, so synthesis is decidable when these two parameters are bounded. For arbitrary partial decoders, however, fixed-code split totality is Π_1^0-complete already for 0.7; every nontrivial valid table is total for that code. All two-element split quotients have period two from heap 1 and admit an explicit finite mask classification; the general valid-table case remains open.*

3. *Every nontrivial finite quotient has a finite-exception unary-heap normal form with tail $H_n \to H_{n-1}$. Its inert pad cannot be retained by a verbatim finite-octal encoding when $|Q| > 2$. In the ordinary subtraction class, the realizable finite quotients are exactly the trivial quotient, $\mathcal{C}_2$, and $\mathcal{T}_n$ for $n \geq 2$. This gives an unbounded-code synthesis algorithm in that class; the code 0.* $\underbrace{33\cdots3}_{2^{n-1}\ \textit{digits}}$ *realizes $\mathcal{T}_n$.*

4. *In misère Grundy's game, the exact quotients generated through heaps 12, 13, and 18 have orders 6, 12, and 24, respectively, with the presentations and reduced translation tables given below. Any finite full Grundy quotient has a heap transition whose option set contains the square of its own value, forcing a nonconstant positive-power cycle. The mate of the canonical representative of H_{13} is not globally equivalent to any Grundy position.*

The normal form in item 3 belongs to a comparison class, while the tame family in the same item is a strict finite-octal theorem. Item 4 combines exact prefixes, a global obstruction, and a conditional structure theorem. None of the four items asserts automatic periodicity for general valid split traces or finiteness of the full Grundy quotient. Sections 2 and 3 build the trace calculus, Section 4 gives the comparison normal form, Section 5 classifies ordinary subtraction realization, and Section 6 gives the Grundy extensions. Section 7 states the formal and open boundaries.

2 Transition tables and deterministic values

Let A be a set of finite impartial games closed under options and disjunctive sum. Under misère play the terminal game is an N-position. For $G, H \in A$, put

$$G \sim_A H \quad \Longleftrightarrow \quad o^-(G + X) = o^-(H + X) \quad \text{for every } X \in A.$$

The quotient $Q(A) = A/\!\sim_A$ is a commutative monoid. It comes with the distinguished subset P of previous-player wins. The pair (Q, P) is *reduced*: if $x \neq y$, then some z has exactly one of xz, yz in P. Also $1 \notin P$, because the terminal position is N.

Plambeck and Siegel developed this quotient method and the generalized mex rule in [7]; Siegel subsequently proved the abstract classification by valid transition tables in [9]. A transition pair is $(x, E) \in Q \times \mathrm{Pow}(Q)$, with product

$$(x, E)(y, F) = (xy, yE \cup xF). \tag{1}$$

A transition table T on (Q, P) is valid when it is complete, closed under the displayed product, parity-correct,

$$(x, E) \in T \quad \Longrightarrow \quad x \in P \iff E \neq \varnothing \text{ and } E \cap P = \varnothing, \tag{2}$$

and well-founded: there is $R : Q \to \mathbb{N}$, with $R(1) = 0$, such that every x has some $(x, E) \in T$ satisfying $R(e) < R(x)$ for all $e \in E$. Siegel's theorem says that a reduced (Q, P) with $1 \notin P$ is an abstract finite misère quotient if and only if it admits such a table.

The natural problem begins after that theorem: a code must generate its table through one translation-invariant heap rule.

2.1 Option-set determinism

The first observation converts the natural problem from a nondeterministic labeling problem into a trace recurrence.

Theorem 2.1 (Option-set determinism)**.** *Let T be a valid transition table on a reduced bipartite monoid (Q, P). If*

$$(x, E) \in T \quad \textit{and} \quad (y, E) \in T,$$

then $x = y$.

Proof. Assume $x \neq y$. Among all contexts separating x and y, choose z with $R(z)$ least. Take a descending pair $(z, F) \in T$. Suppose, after interchanging x, y if necessary, that $xz \in P$ and $yz \notin P$. Closure gives

$$(xz, zE \cup xF), \qquad (yz, zE \cup yF) \in T.$$

Parity for xz says that $zE \cup xF$ is nonempty and contains no element of P. The second option set is nonempty as well: a witness coming from zE is unchanged, while a witness xf is replaced by yf. Since $yz \notin P$, parity therefore gives some P-element in $zE \cup yF$. It cannot lie in zE, which is already contained in the P-free first option set. Hence $yf \in P$ for some $f \in F$, while $xf \notin P$. Thus f also separates x, y, but $R(f) < R(z)$, a contradiction. □

Equivalently, every valid table defines a partial function

$$\mu_T(E) = x \quad \Longleftrightarrow \quad (x, E) \in T. \tag{3}$$

This strengthening of the option-image principle is purely algebraic: it does not assume that T was first obtained from games. The theorem and its least-rank proof are kernel-checked in the Lean module `Ogdoad.MisereTransition`.

We also record the standard local obstruction in a form suited to heap traces. Define the meximal set

$$M_x = \{y \in Q : \text{there is no } z \text{ with } xz, yz \in P\}.$$

Lemma 2.2 (Meximal obstruction)**.** *Every $(x, E) \in T$ satisfies $E \subseteq M_x$.*

Proof. If $e \in E$ and $xz, ez \in P$, choose any table pair (z, F). Closure puts $(xz, zE \cup xF)$ in T, and $ez \in zE$ is a P-valued option of the P-value xz, contradicting parity. □

The fact needed to exclude self-options is stronger than reducedness alone; validity supplies it.

Proposition 2.3 (P-reaching contexts and loop-free records)**.** *Let T be a valid transition table on a nontrivial reduced bipartite monoid (Q, P). For every $x \in Q$ there is a $z \in Q$ with $xz \in P$. Consequently every $(x, E) \in T$ satisfies $x \notin E$.*

Proof. Choose a descending record (x, E). If $E = \varnothing$, determinism against the terminal record $(1, \varnothing)$ gives $x = 1$. Nontrivial reducedness implies $P \neq \varnothing$, so any $p \in P$ is a P-reaching context. Suppose $E \neq \varnothing$. If $x \in P$, take $z = 1$; if $x^2 \in P$, take $z = x$. Otherwise closure puts (x^2, xE) in T. Its option set is nonempty, so parity and $x^2 \notin P$ force $xe \in P$ for some $e \in E$. Thus every value has a P-reaching context.

The meximal obstruction gives $E \subseteq M_x$. If $x \in E$, then $x \in M_x$, contradicting a context z for which both copies of xz lie in P. □

The one-element quotient has only the terminal record and satisfies the same no-self conclusion trivially. Reducedness without a valid table would not be enough for the P-reaching statement.

2.2 The universal star action

Every nontrivial valid quotient contains the same two-element submonoid. This standard fact is useful both for positive trace theorems and for recognizing which purported obstructions do not distinguish a ruleset.

Proposition 2.4 (Free star involution)**.** *In a nontrivial reduced valid table there is an element $a \neq 1$ with*

$$(a, \{1\}) \in T, \qquad a^2 = 1, \qquad aq \neq q \quad (q \in Q).$$

Consequently every nonempty finite ideal, in particular the kernel of a finite quotient, has positive even cardinality.

Proof. Choose a nonidentity element of least rank and a descending record for it. Every option value is 1, and the option set cannot be empty by determinism against $(1, \varnothing)$. This gives $(a, \{1\})$ and $a \in P$. Its square is $(a^2, \{a\})$, so $a^2 \notin P$.

We prove $a^2 z \in P \iff z \in P$ by induction on the rank of a descending record (z, F). If F is empty, then $z = 1$ and both sides are false. Otherwise, if $a^2 z \in P$, its options $a^2 f$ are outside P, hence so are all $f \in F$ by induction; thus $z \in P$. Conversely, if $z \in P$, then $az \notin P$, since the product of $(a, \{1\})$ with (z, F) has option z. Also every $a^2 f$ is outside P by induction. These are exactly the options of the product of $(a^2, \{a\})$ with (z, F), so $a^2 z \in P$. Reducedness now gives $a^2 = 1$.

For any q, multiply its record by $(a, \{1\})$. The resulting record has value aq and option q, so Proposition 2.3 gives $aq \neq q$. Multiplication by a is therefore a fixed-point-free involution on Q and on every ideal. Its orbits have size two. □

Lean checks star existence in the normal-form module, and the involution and free action in `MisereParityTrace`. In particular, a nontrivial kernel is a universal requirement on finite misère quotients, not a special obstruction for Grundy's game.

3 Exact octal traces

3.1 The trace criterion

Write a finite octal code as three masks. Removing k counters may take a whole heap when C_k holds, may leave one nonempty heap when A_k holds, and may leave two nonempty heaps when B_k holds. These are respectively the $1, 2, 4$ bits of the kth octal digit. Let d be the last nonzero digit, put $x_0 = 1$, and seek $x_n = \Phi(H_n)$ for $n \geq 1$. The quotient image of the option set of a heap is exactly

$$\begin{aligned} E_n = \big(\{1\} \text{ if } C_n\big) &\cup \{x_{n-k} : A_k,\ k < n\} \\ &\cup \{x_i x_{n-k-i} : B_k,\ k \leq n-2,\ 1 \leq i \leq (n-k)/2\}. \end{aligned} \tag{4}$$

Equal parts are permitted in an octal split.

Proposition 2.3 immediately gives three absolute exclusions for every exact trace:

$$C_n \Longrightarrow x_n \neq 1, \tag{5}$$

$$A_k,\ k < n \Longrightarrow x_n \neq x_{n-k}, \tag{6}$$

$$B_k,\ i, j \geq 1,\ i + j = n - k \Longrightarrow x_n \neq x_i x_j. \tag{7}$$

These are not bounded-search heuristics: the indicated value occurs in E_n, so equality would make the heap record a forbidden self-option. For example, if the tail has period p, no active one-remainder offset can be a multiple of p; applying the second exclusion sufficiently far into the tail gives an immediate contradiction.

Theorem 3.1 (Exact trace criterion)**.** *Fix a finite octal code Γ and a finite reduced bipartite monoid (Q, P). Then $Q(\Gamma) \cong (Q, P)$ if and only if there are a valid table T on (Q, P) and a sequence $(x_n)_{n\geq 0}$ such that*

1. *$x_0 = 1$;*
2. *$(x_n, E_n) \in T$ for every $n \geq 1$, with E_n given by the displayed octal-trace formula;*
3. *the elements x_n generate Q.*

For fixed T and Γ, the sequence is unique whenever it exists.

Proof. Necessity follows by taking the actual transition algebra and the quotient images of the heaps. Conversely define Φ multiplicatively on heap multisets. The transition pair of any position is the product of its heap pairs. Closure of T therefore puts $(\Phi(G), \Phi''G)$ in T for every position G. Induction on total counter count, using parity, gives

$$o^-(G) = P \quad \Longleftrightarrow \quad \Phi(G) \in P.$$

Generation makes Φ onto. Equal images are indistinguishable by the displayed equivalence; distinct images are distinguished by a context in Q, and surjectivity realizes that context by a position. Thus the fibers are exactly the indistinguishability classes. Uniqueness is Theorem 2.1 applied recursively. □

This theorem identifies the general unresolved core precisely: a finite alphabet quadratic-convolution recurrence must remain defined forever. It also separates exact certificates from bounded signatures.

3.2 Finite-state and periodic traces

Suppose every digit lies in $\{0, 1, 2, 3\}$, so B_k is always false. For $n > d$, whole-heap removal is also inactive, and

$$E_n = \{x_{n-k} : A_k,\ 1 \leq k \leq d\}. \tag{8}$$

Theorem 3.2 (Automatic periodicity)**.** *Let Γ be a finite no-split octal code with last nonzero digit d. If $Q(\Gamma)$ is finite, then its single-heap quotient word $x_n = \Phi(H_n)$ is ultimately periodic. A repeated state occurs among at most $|Q|^d + 1$ consecutive d-windows after the exceptional prefix.*

Proof. The ordered window

$$W_n = (x_{n-d}, \ldots, x_{n-1}) \in Q^d$$

determines E_n by the preceding formula. By Theorem 2.1, E_n determines x_n. Thus W_n determines W_{n+1}. The deterministic map acts on at most $|Q|^d$ states, so one state repeats and the entire subsequent orbit is periodic. □

Corollary 3.3. *For a fixed finite no-split code and a fixed finite reduced (Q, P), exact realization is decidable. The same is true when the code length is bounded in advance.*

Indeed there are finitely many transition tables on Q; validity is a finite condition, and each induced trace either fails or enters a cycle. Generation and parity can then be checked on finite data. The existential problem over codes of unbounded length is not settled by this argument.

Splitting is the exact place where finite memory fails. The new term

$$S_m = \{x_i x_j : i, j \geq 1,\ i + j = m\}$$

depends on an unbounded convolution of the prefix. Nevertheless an already periodic trace remains finitely certifiable.

Proposition 3.4 (Periodic convolution)**.** *Assume $x_{n+p} = x_n$ for all $n \geq N$, where $N, p \geq 1$. Then*

$$S_{m+p} = S_m \qquad (m \geq 2N + p).$$

Consequently the records (x_n, E_n) of a length-d octal code are periodic for all $n \geq 2N + p + d$.

Proof. For a pair $i + j = m + p$, one of i, j is at least $N + p$; subtract p from that component and use periodicity. Conversely, for $i + j = m$, one component is at least N; add p to it. Commutativity permits reordering. The one-remainder terms are easier, and whole-heap bits are inactive past d. □

More explicitly, if the code is supported through digit d, then checking one full period of records after any bound $B \geq 2N + p + d$ certifies every later heap record. Thus a proposed preperiod, period, valid table, finite trace through that first tail period, and generation check form an exact infinite-rule certificate. This is a transition-table form of the periodicity argument in [8, 7].

3.3 Compressing periodic octal witnesses

A periodic pretending function bounds the length of a code realizing it, even when the given realization has arbitrarily large offsets. The transition records themselves need not be preserved. We first make precise the enlargement of a valid table that this argument uses.

Lemma 3.5 (Parity-preserving meximal saturation)**.** *Let T be a valid transition table on a reduced bipartite monoid (Q, P). Let $\widehat{T}$ consist of all pairs (x, E) such that*

$$E \subseteq M_x, \qquad x \in P \iff E \neq \varnothing \text{ and } E \cap P = \varnothing,$$

and some $(x, F) \in T$ satisfies $F \subseteq E$. Then $\widehat{T}$ is a valid transition table containing T.

Proof. The meximal obstruction gives $T \subseteq \widehat{T}$, so completeness and the descending witnesses are inherited. It remains to prove closure. Take $(x, E), (y, G) \in \widehat{T}$ with baseline records $(x, F), (y, H) \in T$, where $F \subseteq E$ and $H \subseteq G$. Their product has option set $K = yE \cup xG$, containing the baseline option set $K_0 = yF \cup xH$ of a record $(xy, K_0) \in T$. Moreover $K \subseteq M_{xy}$: if $e \in E$ and both xyz, yez belonged to P, then the context yz would contradict $e \in M_x$; the other summand is symmetric.

If $K_0 \neq \varnothing$, then $K \neq \varnothing$. When $xy \in P$, meximal safety excludes P-elements of K by taking context 1. When $xy \notin P$, parity of the baseline record supplies a P-element of $K_0 \subseteq K$. Thus the product is parity-correct. If $K_0 = \varnothing$, then $F = H = \varnothing$, and option-set determinism against $(1, \varnothing)$ gives $x = y = 1$. Now $K = E \cup G$. If this union is nonempty, one factor has a nonempty option set and its individual parity supplies a P-element; if it is empty, the product is the terminal record. Since $1 \notin P$, both cases are parity-correct. Hence the product belongs to $\widehat{T}$. □

The individual parity condition cannot be omitted when a baseline is empty. It is exactly the condition handled separately for terminal heaps in the following theorem.

Theorem 3.6 (Periodic octal witness compression)**.** *Suppose a finite octal code has exact reduced quotient (Q, P) and heap pretending function $(x_n)_{n\geq 0}$. If $N, p \geq 1$ and $x_{n+p} = x_n$ for every $n \geq N$, then the same sequence is the exact pretending function of a finite octal code whose last nonzero digit is at most*

$$B = 3N + 2p.$$

Proof. Keep all original bits at offsets at most B and discard all bits at larger offsets. For every original one-remainder or split bit at offset $k \geq N + p$, also set that same bit at the unique offset

$$\rho(k) \in \{N, N+1, \ldots, N+p-1\}, \qquad \rho(k) \equiv k \pmod{p}.$$

This is done for every such original persistent bit, including those with $k \leq B$. Whole-heap bits are not copied. For original persistent offsets $k < N + p$, write $\rho(k) = k$; these bits were already retained. Every original persistent bit therefore has a representative at an offset at most $N+p-1$.

Let E_n and E'_n be the original and new option-value sets, both computed from the given sequence x, and let T be the original valid transition table. Every option from a retained bit lies in $E_n \subseteq M_{x_n}$. An added bit at $k' = \rho(k)$ comes from an original offset $k \geq k'$ with $k - k' = tp$ for some $t \geq 0$. A new move from heap n using this bit has exactly the same remainder heap or heaps as the original move from heap $n + tp$ using offset k. Its existence implies $n > k' \geq N$, so $x_{n+tp} = x_n$. Thus every added option is also mexiMal-safe, and

$$E'_n \subseteq M_{x_n} \qquad (n \geq 0).$$

If Q is trivial, validity allows only the record $(1, \varnothing)$. The original code therefore has no active bits, and there is nothing to prove. Assume henceforth that Q is nontrivial, and let a be its universal star value. In particular, $(a, \{1\}) \in T$, $a \in P$, $a^2 = 1$, and $(1, \{a\}) \in T$.

For $n \leq B$, every original move uses an offset at most B, so $E_n \subseteq E'_n$. If $E_n \neq \varnothing$, the new record is parity-correct: a P-value has no P-option by mexiMal safety, while an N-value retains a P-option from its nonempty baseline. Therefore $(x_n, E'_n) \in \widehat{T}$. If $E_n = \varnothing$, then the original heap n is terminal and $x_n = 1$. No persistent bit is active at any smaller heap: a one-remainder bit active below n is also active at n, and the same holds for a split bit. Every smaller original heap consequently has either no options or only the empty position as an option, so its value lies in $\{1, a\}$. Every new option at n, including a product from two remainder heaps, therefore has value in $\{1, a\}$. But $1 \notin M_1$, as witnessed by the context $a \in P$. MeximaL safety forces $E'_n = \varnothing$ or $E'_n = \{a\}$. Both resulting records already belong to T, which resolves the terminal case without assuming generic saturation preserves parity.

If the original code has no persistent bits, its sequence is eventually 1; periodicity from N then implies $x_n = 1$ for every $n \geq N$. Every active whole-heap bit is therefore below N, since such a bit gives heap value a. The construction leaves this code unchanged, proving the theorem in this case. Now assume some persistent bit is present.

Write

$$F_m = \{x_i x_j : i, j \geq 1,\ i + j = m\}.$$

For $m \geq 2N + p$ one has $F_{m+p} = F_m$. Indeed, in a pair summing to m, some part is at least N and can be increased by p; in a pair summing to $m + p \geq 2N + 2p$, some part is at least $N + p$ and can be decreased by p. Periodicity preserves the corresponding products.

Fix $n > B$, and choose $m \equiv n \pmod{p}$ sufficiently large that it is beyond the last original digit and that, for every original persistent offset k, the residual $m - k$ is at least $2N + p$. The original

whole-heap contribution at m is absent, and E_m is nonempty because some persistent bit is active. Every representative satisfies

$$n - \rho(k) \geq (B+1) - (N+p-1) = 2N + p + 2.$$

Thus $m - k$ and $n - \rho(k)$ are congruent modulo p and lie in the stable range. Their one-remainder values agree, and their split convolution sets agree. Every original contribution to E_m is therefore present in E'_n. We have obtained a nonempty baseline

$$(x_n, E_m) = (x_m, E_m) \in T, \qquad E_m \subseteq E'_n.$$

Meximal safety and this nonempty baseline give individual parity as before, hence $(x_n, E'_n) \in \widehat{T}$ for every $n > B$ as well.

All new heap records now lie in the valid table $\widehat{T}$. The exact trace criterion proves that their sequence is an outcome map for the new ruleset. Its values still generate Q, because the sequence was unchanged; reducedness therefore gives the same exact quotient. □

Corollary 3.7 (Bounded periodic-complexity synthesis)**.** *For a given finite reduced bipartite monoid (Q, P) and integers $N, p \geq 1$, it is decidable whether some finite octal code realizes (Q, P) with a pretending function periodic from N with period p. The same holds when N and p are upper bounds rather than specified parameters.*

Proof. There are finitely many Q-valued prefix-and-period words for the specified N, p, with $x_0 = 1$. By Theorem 3.6, any realizable word has a realizing code of length at most $3N + 2p$. Enumerate those finitely many words and codes, together with the finitely many transition tables on Q. Check table validity, generation of Q, and the finite periodic trace certificate. These are finite decisions and the exact trace criterion proves both directions. For upper bounds, enumerate the finitely many admissible parameter pairs. □

The Lean module `MisereOctalCompression` checks the actual mask transformation, its $3N+2p$ bound, and every compressed heap record's membership in the valid saturated table. It uses the saturation and terminal-prefix modules; the quotient interpretation and finite enumeration are the paper-level composition of these certificates.

This reduces code-length control, within the periodic branch of the realization problem, to control of the preperiod and period. It supplies no computable bound for those two parameters in terms of $|Q|$, and it does not prove that a splitting code with finite quotient has an ultimately periodic pretending function.

3.4 The Boolean-language boundary

The convolution obstruction can be made exact. For languages $L, M \subseteq \mathbb{N}$, write

$$L + M = \{i + j : i \in L,\ j \in M\}.$$

Fix a candidate word $x = (x_n)$ and put $X_q = \{n : x_n = q\}$. For each $q \in Q$, let A_q be the language of heap sizes at which q occurs as an option value. In terms of the three masks it is

$$A_q = C_q \ \cup \bigcup_{k \in A} \big(k + (X_q \cap \mathbb{N}_{>0})\big) \ \cup \bigcup_{k \in B} \bigcup_{uv = q} \big(k + (X_u \cap \mathbb{N}_{>0}) + (X_v \cap \mathbb{N}_{>0})\big), \tag{9}$$

where $C_1 = C$ is the finite whole-removal language and $C_q = \varnothing$ for $q \neq 1$. Bounds excluding empty remainders are understood in the shifted terms. For $S \subseteq Q$, define the exact Boolean cell

$$D_S = \bigcap_{q \in S} A_q \ \cap \bigcap_{q \notin S} (\mathbb{N} \setminus A_q). \tag{10}$$

Proposition 3.8 (Exact language equations)**.** *Let $\mu : \mathrm{Pow}(Q) \rightharpoonup Q$ be any finite partial decoder. A word x is its total octal trace exactly when*

$$X_q = \bigcup_{\mu(S)=q} D_S \qquad (q \in Q) \tag{11}$$

and the failure language

$$F_\mu = \bigcup_{\mu(S)\ \mathit{undefined}} D_S \tag{12}$$

is empty. Coefficient n belongs to F_μ exactly when the decoder has no value on the complete option set at n.

Proof. The first displayed equation is simply the whole-, one-, and two-remainder definition sorted by quotient value. Hence $n \in D_S$ if and only if the complete option-value set at H_n is exactly S. Applying μ coefficientwise gives both displayed conclusions. □

These equations are unary, but they are not finite-state equations. Unary conjunctive grammars already generate nonregular languages such as $\{a^{4^r} : r \geq 0\}$ [5]. In fact the unconstrained trace problem reaches the full effective boundary.

Theorem 3.9 (Arbitrary-decoder hardness)**.** *For the single fixed octal code* 0.7*, the following problem is Π^0_1-complete: given a finite commutative monoid M and an arbitrary finite partial decoder $\mu : \mathrm{Pow}(M) \rightharpoonup M$, decide whether the trace starting from $x_0 = 1$ is total.*

Proof. Non-totality has a least failing coefficient, so it is recursively enumerable and totality is in Π^0_1.

For hardness, start with a unary unambiguous conjunctive grammar. Emptiness for these grammars is undecidable by an effective reduction from Turing machine emptiness [6, Lemma 12]. Effectively put the grammar in epsilon-free binary conjunctive normal form and prefix a fixed two-letter word, so that lengths 0 and 1 are irrelevant while emptiness is preserved. Unambiguity is needed only for the cited source theorem; the normalization need not preserve it. Let $\mathcal{N}$ be the resulting finite nonterminal set and let $V_n \subseteq \mathcal{N}$ be its truth vector on a^n. Take the full, effectively known Boolean cube $C = \mathrm{Pow}(\mathcal{N})$ as the color set and form the commutative monoid

$$M_C = \{\text{multisets on } C \text{ of degree at most } 2\} \sqcup \{\bot\}.$$

Multiplication is multiset union until degree 2 and is $\bot$ after overflow, with $\bot$ absorbing; the empty multiset is the identity. This is associative: for three ordinary factors both parenthesizations give their union when its total degree is at most two and $\bot$ otherwise, while an existing $\bot$ remains absorbing.

Assign H_h the singleton marker $[V_{h+1}]$. Under 0.7, the option set of H_h for $h \geq 3$ is exactly

$$\{[V_h]\} \ \cup \{[V_{i+1}, V_{j+1}] : i, j \geq 1,\ i + j = h - 1\}. \tag{13}$$

The degree-two markers encode every binary cut of the word of length $h+1$ whose two parts have length at least two. The degree-one predecessor marker, together with the fixed vector V_1, encodes the two cuts having a part of length one. Because the alphabet is unary, unordered pairs lose no cut information. Thus the exact option set determines V_{h+1}.

More explicitly, the normalized grammar gives an effective finite update map

$$\mathsf{H} : C \times \mathrm{Pow}(\{[u, v] : u, v \in C\}) \longrightarrow C.$$

It computes the next truth vector from the predecessor vector and the set of unordered binary-cut vectors; the two cuts involving V_1 are folded into the update. For every decodable signature

$$S = \{[c]\} \cup R, \qquad R \subseteq \{[u, v] : u, v \in C\},$$

define $\mu(S) = [\mathsf{H}(c, R)]$ when the start nonterminal is absent from $\mathsf{H}(c, R)$, and leave $\mu(S)$ undefined when it is present. This is a uniform finite definition, independent of which signatures the eventual trace reaches. Malformed signatures may be left undefined. The genuinely exceptional H_1 cell computes V_2 directly. The H_2 cell is the decodable signature $(c, R) = (V_2, \varnothing)$, for which $\mathsf{H}(V_2, \varnothing) = V_3$ because only the two length-one cuts occur. Both cells are left undefined exactly when their output start bit is true.

Before the first accepted length, induction now gives precisely the asserted markers. At the first accepted length the decoder fails. Hence the trace is total if and only if the grammar language is empty. Turing-machine emptiness is Π^0_1-complete, completing the reduction. □

The decoder in Theorem 3.9 is deliberately arbitrary. It need not extend to a reduced, closed, parity-correct, ranked misère transition table. Therefore the theorem neither proves undecidability of natural realization nor decides a fixed valid-table trace. Its truncated monoid also has an absorbing element $\perp$, which cannot occur in any nontrivial valid quotient: Proposition 2.4 would give $a\perp \neq \perp$. The hardness theorem proves the more surgical statement needed here: any decision argument must use the valid-table axioms in a load-bearing way. Finiteness of Q, commutativity of its multiplication, and unary input alone are insufficient.

3.5 What validity forces on the hardness code

The code in Theorem 3.9 becomes completely tractable when its decoder comes from a valid table. The following statement permits split moves and does not assume a finite table.

Theorem 3.10 (Odd-removal traces)**.** *Suppose all active removal offsets of an octal code are odd, and its first digit has both the whole-removal and one-remainder bits. In every nontrivial reduced valid table the unique total trace is*

$$x_n = a^n, \qquad E_n = \{a^{n-1}\} \quad (n \geq 1),$$

where a is the star value of Proposition 2.4. The exact quotient is $\mathcal{C}_2 = (\langle a \mid a^2 = 1\rangle, \{a\})$. In particular, a valid-table trace for 0.7 *exists if and only if its ambient monoid is nontrivial; this is a totality statement even when the trace does not generate that ambient monoid.*

Proof. A move removes an odd number k of counters. With $x_n = a^n$, a one-heap option and a split option both have value $a^{n-k} = a^{n-1}$. A whole-removal move occurs at odd n and has value $1 = a^{n-1}$. The first-digit hypotheses ensure at least one option at every positive heap size, so the displayed option-set equality is exact.

Closure contains every positive power of $(a, \{1\})$, whose nth power is $(a^n, \{a^{n-1}\})$. Thus all heap records lie in the table, and uniqueness follows from determinism. On the generated submonoid $\{1, a\}$ the P-portion is $\{a\}$, and these two elements are separated by the empty context, proving the exact quotient assertion. For a one-element ambient monoid, the terminal record forces $1 \notin P$, whereas the first 0.7 heap record would be $(1, \{1\})$, contradicting parity. This proves the final equivalence. □

Thus the arbitrary-decoder hardness carrier itself supplies no hard valid instance. The Lean module `MisereParityTrace` checks the option identity, totality, trace uniqueness, and period two.

3.6 Complete two-element split classification

There is also an exact classification with no restriction on the removal parities in advance. Write $C, A, B \subset \mathbb{N}_{>0}$ for the whole-, one-, and two-remainder masks, and assume $B \neq \varnothing$. Define the first heap size at which a persistent move is available by

$$L = \min\big(\{k+1 : k \in A\} \cup \{k+2 : k \in B\}\big).$$

All arithmetic on the bits u, v in the next theorem is in $\mathbb{F}_2$.

Theorem 3.11 (Two-element split quotients)**.** *A finite octal code with $B \neq \varnothing$ has exact quotient $\mathcal{C}_2$ if and only if there are bits $(u, v) \neq (0, 0)$ satisfying*

$$\begin{aligned} uk &= 1 \quad (k \in A), \\ uk + v &= 1 \quad (k \in B \cup C), \\ \{n : 1 \le n < L,\ un + v = 1\} &\subseteq C. \end{aligned}$$

Its heap map is then $x_n = a^{un+v}$ for every $n \ge 1$. There are only three candidates: the constant word a, the word a^n, and the word a^{n+1}. Hence these finite mask tests decide every split-code instance with target $\mathcal{C}_2$.

Proof. In a two-element valid table, no-self and parity allow exactly the records

$$(1, \varnothing), \qquad (a, \{1\}), \qquad (1, \{a\}).$$

Fix one active split offset k. Every split option of H_{k+i+j} has the same value. Comparing the pairs $(1, 3)$ and $(2, 2)$ at total 4 gives $x_1 x_3 = x_2^2 = 1$, hence $x_3 = x_1$. Comparing $(1, n+2)$ and $(3, n)$ now gives $x_1 x_{n+2} = x_3 x_n$, hence $x_{n+2} = x_n$ for every $n \ge 1$. Each pair is a legal octal split; equal parts are allowed. The positive heap word therefore has the form $x_n = a^{un+v}$. The constant word 1 is impossible because an active split would be a self-option.

A one-remainder move changes the exponent by uk, so no-self is exactly the first condition. A split option has exponent $u(n-k)$, so no-self is $uk + v = 1$. A whole-removal option has value 1, requiring $un + v = 1$ at its source $n \in C$. These give the second condition. Below L, only whole-removal moves are possible; a heap there with value a must have such a move, giving the third condition.

Conversely, these conditions make every available option the complement of the proposed heap value. Every heap $n \ge L$ has a persistent move; for $n < L$, the second and third conditions say that a heap is terminal exactly when its proposed value is 1. Thus all records belong to the displayed three-record table. This table is valid, and each of the three candidate words attains a, so generation and the trace criterion prove sufficiency. □

For example, 0.51 has constant heap value a, 0.5 has heap value a^n, and 0.05 has heap value a^{n+1}. These examples also show why the first-digit hypotheses of Theorem 3.10 are sufficient rather than necessary. The period-two rigidity is kernel-checked in `MisereParityTrace`; the explicit mask classification above is its paper-level algebraic consequence.

4 A finite-exception heap normal form

The abstract table theorem does admit a uniform one-species heap realization once the finite prefix is allowed to be source-local. This comparison result pinpoints why the direct construction does not immediately become octal.

Theorem 4.1 (Finite-exception unary-heap normal form)**.** *Let (Q, P) be a nontrivial finite reduced bipartite commutative monoid. Suppose T is a closed parity-correct transition table and $R : Q \to \mathbb{N}$, with $R(1) = 0$, such that every $q \in Q$ has a record $(q, E_q) \in T$ satisfying $R(e) < R(q)$ for all $e \in E_q$. Then there are N and a finite-description impartial ruleset on numerical heaps $H_1, H_2, \ldots$ such that*

1. *for $n \le N$, every option of H_n is a single heap H_j with $j < n$;*
2. *H_N is terminal;*
3. *for every $n > N$, $\mathrm{opts}(H_n) = \{H_{n-1}\}$; and*
4. *the exact misère quotient of all finite sums of these heaps is isomorphic to (Q, P).*

Proof. First find a star record. Choose a nonidentity value a of least rank and a descending record (a, E). Every member of E has smaller rank, hence is 1. The set E cannot be empty, since determinism against $(1, \varnothing)$ would give $a = 1$. Therefore $(a, \{1\}) \in T$.

For each $q \in Q$, choose a descending record (q, E_q). Enumerate the elements of Q in nondecreasing rank and make one prefix heap B_q with options B_e for $e \in E_q$. Every target is an earlier heap. Append one inert padding heap I, followed by heaps $U_0, U_1, \ldots$ with

$$\mathrm{opts}(U_0) = \{I\}, \qquad \mathrm{opts}(U_{k+1}) = \{U_k\}.$$

Assign the values

$$\Phi(B_q) = q, \qquad \Phi(I) = 1, \qquad \Phi(U_k) = a^{k+1}.$$

Closure of T and $(a, \{1\}) \in T$ give by induction

$$(a^{k+1}, \{a^k\}) \in T,$$

so every single-heap record lies in T. The transition record of a sum is the product of its component records and therefore also lies in T. All moves strictly lower the sum of the displayed heap ranks. Induction on that rank and parity show

$$o^-(G) = P \quad \Longleftrightarrow \quad \Phi(G) \in P$$

for every position G. Each q is represented by B_q, so Φ is surjective. Equal values are indistinguishable, while reducedness and surjectivity realize a separating context for unequal values. Hence the fibers of Φ are exactly the indistinguishability classes. □

This is an eventually translation-invariant finite-exception heap normal form, not a finite-octal realization. Turning a prefix edge $H_i \to H_{i-d}$ into an octal digit at offset d also creates translated moves from every larger heap. The construction supplies no values for those additional targets. Controlling that cross-talk—possibly by a different, generator-level encoding—remains an unresolved bridge. The comparison class and finite octal games are therefore not ordered by inclusion: the former permits arbitrary source-local prefix moves but only a unary tail, whereas the latter permits persistent multi-removal and split moves.

The inert pad is not merely an expositional nuisance.

Proposition 4.2 (Inert-pad obstruction)**.** *Assume $|Q| > 2$, and choose one strictly descending record (q, E_q) for each $q \in Q$ as in Theorem 4.1. There is no finite octal code with prefix heaps B_q followed by a later terminal heap I such that every B_q has exact option-value set E_q.*

Proof. Determinism identifies every chosen empty record with $(1, \varnothing)$ and identifies all chosen records with option set $\{1\}$ with the unique star value. Since $|Q| > 2$, some chosen E_q therefore contains an $e \neq 1$.

A whole-heap-removal bit contributes only 1, so the option e at B_q must come from a one-remainder or two-remainder bit at some removal size k. Such a bit is persistent. If it leaves one remainder at the earlier heap B_q, then $k < |B_q| < |I|$ and it leaves a remainder at I. If it leaves two remainders at B_q, then $k + 2 \leq |B_q| < |I|$ and it also gives a split option at I. Either conclusion contradicts that I is terminal. □

Thus no placement trick, including spacing the representatives farther apart, turns the proof of Theorem 4.1 verbatim into an octal proof. This does not rule out octal universality: a successful compiler may replace the inert pad by an active bridge or may encode quotient generators contextually rather than representing every chosen descending record by one heap.

5 Complete realization for ordinary subtraction

An ordinary subtraction game has a finite set $S \subset \mathbb{N}_{>0}$ of allowed removal sizes; its octal digits are 3 on S and 0 elsewhere. Unlike a general no-split code, it always permits taking the whole heap whenever it permits the corresponding subtraction. This subclass admits a complete realization classification, including synthesis over unbounded code lengths. The underlying zero/one mechanism is Ferguson's classical theorem [2]; see also [4, Section 6.2]. We give the proof to make the quotient construction self-contained.

5.1 The zero/one shift and the full outcome law

Let $g(n) = \operatorname{mex}\{g(n-t) : t \in S,\ t \leq n\}$ be the normal Sprague–Grundy sequence, with $g(0) = 0$. Suppose $S \neq \varnothing$ and put $s = \min S$.

Lemma 5.1 (Ferguson's shift)**.** *For every $n \geq 0$,*

$$g(n) = 0 \quad \Longleftrightarrow \quad g(n+s) = 1.$$

Every nonterminal heap of value zero has an option of value one.

Proof. Use strong induction on n. If $g(n) = 0$, the legal subtraction s shows $g(n+s) \neq 0$. If also $g(n+s) \neq 1$, mex gives an option $m = n + s - t$ with $g(m) = 1$. Since heaps below s are terminal, $m \geq s$. Induction at $m - s = n - t < n$ then gives $g(n-t) = 0$, contradicting the mex condition at n. Conversely, if $g(n+s) = 1$ but $g(n) > 0$, choose a legal t with $g(n-t) = 0$. Induction gives $g(n+s-t) = 1$, contradicting the mex condition at $n+s$.

For the last assertion, a nonterminal zero-valued n satisfies $n \geq s$ and $g(n-s) \neq 0$. Choose $t \in S$ with $t \leq n - s$ and $g(n-s-t) = 0$. The shift theorem gives $g(n-t) = 1$, and t is legal at n. □

Theorem 5.2 (Subtraction outcome law)**.** *For any finite sum of heaps, put $v_i = g(h_i)$ and $w = \bigoplus_i v_i$. Its misère outcome is P exactly when*

$$\begin{cases} w = 1, & \text{every } v_i \leq 1, \\ w = 0, & \text{some } v_i \geq 2. \end{cases}$$

Proof. Induct on total counters. Every move changes exactly one v_i to a unequal value. If all values are small and their parity is odd, a move to another small value changes that parity, while a move to a large value makes the XOR nonzero. If a large value occurs and $w = 0$, at least two large values occur. After any move one remains and the XOR is nonzero. Thus no listed P-position has a listed P-option.

Conversely, consider a nonterminal unlisted position. With all values small and parity even, any value-one component can move to value zero by mex. If no such component exists, some zero-valued component is movable, and Lemma 5.1 gives a move to value one. If exactly one large value occurs, move it to zero or one to make the final parity of the small values odd; both targets occur by mex. If at least two large values occur, the ordinary Nim argument selects a component v with $v \,\mathsf{xor}\, w < v$. Mex supplies this move, the resulting XOR is zero, and at least one large component remains. Every nonterminal unlisted position therefore has a listed option. Terminal positions have all values zero and are correctly unlisted. □

The module `MisereSubtraction` constructs the actual mex sequence, defines legal heap moves and the literal misère recursion, and checks both results for every heap size and every finite list of heaps. This is an outcome theorem, not verification of a finite table of samples.

5.2 Exact quotients and a construction algorithm

For $r \geq 2$, let

$$Q_r = \{1, a\} \sqcup \{z_v : v \in \mathbb{F}_2^r\}, \qquad P_r = \{a, z_0\},$$

with multiplication

$$a^2 = 1, \qquad z_u z_v = z_{u \mathsf{xor} v}, \qquad a z_v = z_{v \mathsf{xor} 1}. \tag{14}$$

This is the tame quotient $\mathcal{T}_r$, of order $2^r + 2$ [7, 8]. Also write $\mathcal{I}$ for the one-element quotient with empty P-portion.

Theorem 5.3 (Complete subtraction classification)**.** *The exact misère quotients of finite ordinary subtraction games are precisely*

$$\mathcal{I}, \qquad \mathcal{C}_2, \qquad \mathcal{T}_r \ (r \geq 2).$$

For $S \neq \varnothing$, let $M = \max_{n \geq 0} g(n)$. If $M = 1$, the quotient is $\mathcal{C}_2$. Otherwise it is $\mathcal{T}_r$, where $2^{r-1} \leq M < 2^r$, with heap map

$$\Phi(H_n) = \begin{cases} 1, & g(n) = 0, \\ a, & g(n) = 1, \\ z_{g(n)}, & g(n) \geq 2. \end{cases}$$

Both the exact quotient of a given subtraction code and realization of a given finite reduced bipartite monoid by some subtraction code are decidable.

Proof. Each heap has at most $|S|$ options, so $g(n) \leq |S|$. The maximum M therefore exists, and every value $0, \ldots, M$ occurs: a heap attaining M has options attaining every smaller value. Also $M \geq 1$ since $g(s) = 1$. If $M = 1$, Theorem 5.2 gives exactly the two parity classes. If $M \geq 2$, the same theorem says that the multiplicative map displayed above has the correct P-preimage.

This map is onto. All SG values through 2^{r-1} occur, so its image contains

$$a, z_2, z_4, \ldots, z_{2^{r-1}}, \qquad z_0 = z_2^2, \qquad z_1 = a z_0.$$

These generate the whole kernel. Reducedness is explicit: distinct z_u, z_v are separated by z_u; $1, a$ by 1; $1, z_v$ by a unless $v = 1$, when z_1 works; and a, z_v by 1 unless $v = 0$, when z_0 works. Thus the map's fibers are exactly indistinguishability. The empty subtraction set gives $\mathcal{I}$.

For a given nonempty S, put $d = \max S$. The next SG value is determined by the preceding d values, each in $\{0, \ldots, |S|\}$. A repeated window therefore occurs among $(|S| + 1)^d + 1$ successive windows starting at $(g(0), \ldots, g(d-1))$. The finite prefix and ensuing cycle determine M and the complete heap map.

For synthesis, test the input finite multiplication table and P-portion for isomorphism with the listed candidates; its cardinality leaves only finitely many possibilities. The empty code and 0.3 realize $\mathcal{I}$ and $\mathcal{C}_2$. Corollary 5.4 below supplies a code for each $\mathcal{T}_r$. Failure of these finite isomorphism tests rules out every ordinary subtraction code, regardless of its length. □

Corollary 5.4 (Uniform tame-family realization)**.** *For every $r \geq 2$, the code with 2^{r-1} consecutive digits 3 has exact quotient $\mathcal{T}_r$.*

Proof. Put $m = 2^{r-1}$ and take $S = \{1, \ldots, m\}$. Direct mex induction gives $g(n) = n \bmod (m+1)$: before $m + 1$ all smaller residues occur, and afterward the preceding m terms contain every other residue. Thus $M = m$, and Theorem 5.3 applies. □

This classifies all ordinary subtraction codes, not all no-split codes. Digits 1 and 2 decouple whole removal from leaving a remainder and can produce wild behavior. In Siegel's finite $|P| = 2$ classification, the remaining arm comprises tame extensions of $\mathcal{R}_8$ [9]; its base is realized by 0.75 [7]. No uniform finite-octal realization of all those extensions is established here.

6 Exact quotients and constraints for misère Grundy's game

In Grundy's game a heap may be split into two unequal positive heaps:

$$\operatorname{opts}(H_n) = \{H_i + H_{n-i} : 1 \leq i < n - i\}.$$

The rule preserves total counters but decreases the well-founded quantity $\sum_h \max(h-2, 0)$. Its misère form is a longstanding difficult example; the early analysis goes back to Grundy and Smith [3] and is discussed in [1].

6.1 The tame prefix through heap 12

We first give an exact transition certificate through heap 12. Write

$$\mathcal{T}_2 = \langle a, b \mid a^2 = 1,\ b^3 = b \rangle, \qquad P = \{a, b^2\}.$$

The heap values are

$$x_1 = x_2 = 1, \qquad x_n = \begin{cases} a, & n \equiv 0 \pmod 3, \\ 1, & n \equiv 1 \pmod 3, \\ b, & n \equiv 2 \pmod 3 \end{cases} \quad (3 \leq n \leq 12).$$

Their distinct transition records are

$$(1, \varnothing),\ (a, \{1\}),\ (1, \{a\}),\ (b, \{1, a\}),$$
$$(a, \{1, b\}),\ (1, \{a, b\}),\ (b, \{1, a, ab\}).$$

Closing these records under the transition product gives the following 24 pairs; each row lists the option sets paired with its left-hand value.

1	$\varnothing$; $\{a\}$; $\{a,b\}$; $\{a,ab\}$; $\{a,b,ab\}$
b	$\{1,a\}$; $\{b^2,ab^2\}$; $\{1,a,ab\}$; $\{b^2,ab,ab^2\}$; $\{1,b^2,a,ab\}$; $\{1,a,ab,ab^2\}$; $\{1,b^2,a,ab,ab^2\}$
b^2	$\{b,ab\}$; $\{b,ab,ab^2\}$
a	$\{1\}$; $\{1,b\}$; $\{1,ab\}$; $\{1,b,ab\}$
ab	$\{1,b,a\}$; $\{b,b^2,ab^2\}$; $\{1,b,b^2,a\}$; $\{1,b,a,ab^2\}$; $\{1,b,b^2,a,ab^2\}$
ab^2	$\{b,b^2,ab\}$

The table is closed and every pair obeys parity; a, b occur as heap values and $\mathcal{T}_2$ is reduced. The trace criterion therefore proves:

Proposition 6.1. *The exact misère quotient generated by $H_1, \ldots, H_{12}$ is $\mathcal{T}_2$.*

6.2 The first extension at heap 13

Before extending this quotient, we isolate a substitution point that is easy to miss. Write $G \equiv_{-} H$ when G and H have the same misère outcome after addition of every finite impartial game, not merely every game in the present heap universe.

Lemma 6.2 (Global substitution)**.** *Let U be the additive closure of games G_i, and suppose that each G_i is globally equivalent to a representative R_i lying in an additively closed subuniverse $V \subseteq U$. Then inclusion induces an isomorphism $Q(V) \cong Q(U)$.*

Proof. Global equivalence is an additive congruence, so every $X \in U$ is globally equivalent to the sum $r(X) \in V$ obtained by replacing its components. If $A, B \in V$ agree against all V-contexts, then for every $X \in U$,

$$o^-(A + X) = o^-(A + r(X)) = o^-(B + r(X)) = o^-(B + X).$$

The converse follows because $V \subseteq U$, and every U-class has a representative in V. □

The Grundy–Smith canonical reduction rules are global reductions. Applied to the explicit option lists on heaps through 12, they give

$$\begin{array}{r|l} 0 & H_1, H_2, H_4, H_7, H_{10} \\ a = *1 & H_3, H_6, H_9, H_{12} \\ b = *2 & H_5, H_8, H_{11}. \end{array}$$

These are the reductions displayed in [1, pp. 419–421]. Consequently adjoining H_{13} cannot secretly refine one of these earlier identifications: they remain valid against the new context H_{13} and all of its sums.

Heap 13 has option-value set $\{a, b, b^2\}$. The alleged continuation $x_{13} = 1$ is impossible: multiplying its transition by two copies of the H_5 transition produces

$$(b^2, \{b, b^2, ab, ab^2\}),$$

which violates parity because both the value and one option lie in P. Concretely $H_4 + 2H_5$ is P, while $H_{13} + 2H_5$ has the P-option $H_8 + 3H_5$ and is N.

The extension can still be solved symbolically. Let c denote the new heap value, with transition $c \to \{a, b, b^2\}$. For raw monomials $a^e b^j c^k$, direct induction on (k, j, e) gives the following complete outcome law:

$$a^e b^j c^k \in P \iff \begin{cases} k \text{ even},\ e \text{ even},\ j > 0 \text{ even}; & \text{or} \\ k \text{ even},\ e \text{ odd},\ j = 0; & \text{or} \\ k \text{ odd},\ e \text{ odd},\ j = 0 \text{ or } j \geq 3 \text{ odd}. \end{cases} \tag{15}$$

For completeness, the induction uses only

$$a \to 1, \qquad b \to \{1, a\}, \qquad c \to \{a, b, b^2\}.$$

Here is the complete case induction. Put $\epsilon = e \bmod 2$. For a listed P-state with k even, either $(\epsilon, j) = (0, \text{positive even})$ or $(1, 0)$; the a, b, c moves land respectively in the complementary parity classes, and the three c replacements land at $(1, j), (0, j+1), (0, j+2)$ with k odd. None is listed. For a listed state with k odd, $\epsilon = 1$ and $j = 0$ or odd at least 3; the a move sets $\epsilon = 0$, the b moves make j even, and the c moves land at $(0, j), (1, j+1), (1, j+2)$ with k even. Again none is listed.

Conversely, every unlisted nonterminal state has the following listed option. When k is even:

ϵ	j	move to a P-state
0	0	$c \to a$ (unless $k = 0$, when the position is terminal)
0	1	$b \to a$
0	odd ≥ 3	$b \to 1$
1	positive even	$a \to 1$
1	1	$b \to 1$
1	odd ≥ 3	$b \to a$

When k is odd, the witnesses are

ϵ	j	move	ϵ	j	move
0	0	$c \to a$	0	positive even	$c \to b^2$
0	odd	$c \to b$	1	1	$b \to 1$
1	2	$c \to a$	1	even ≥ 4	$b \to 1$

The omitted odd-k cases are exactly the listed P cases. This proves the misère recursion for all multiplicities; no exponent cutoff is used.

The outcome law also proves the presentation rather than merely suggesting it. It first factors raw monomials through the sixteen forms $a^\epsilon b^j c^\eta$, with $\epsilon, \eta \in \{0, 1\}$ and $j \in \{0, 1, 2, 3\}$, using $a^2 = c^2 = 1$ and $b^4 = b^2$. It is invariant under $b^2 \sim ab^3c$. The congruence generated by this relation pairs the eight forms in the b^2-ideal into four two-element orbits and leaves the eight forms with $j = 0, 1$ fixed. Thus it has exactly twelve candidate classes. The resulting quotient has presentation

$$Q_{13} = \langle a, b, c \mid a^2 = c^2 = 1,\ b^4 = b^2,\ b^2 = ab^3c \rangle, \qquad P_{13} = \{a, ac, b^2\}. \tag{16}$$

It has the twelve normal forms

$$1, c, b, bc, b^2, b^2c, b^3, b^3c, a, ac, ab, abc.$$

Reduction is visible from their distinct P-translation rows:

1	$\{b^2, a, ac\}$	c	$\{b^2c, a, ac\}$
b	$\{b, b^3\}$	bc	$\{bc, b^3c\}$
b^2	$\{1, b^2, abc\}$	b^2c	$\{c, b^2c, ab\}$
b^3	$\{b, b^3, ac\}$	b^3c	$\{bc, b^3c, a\}$
a	$\{1, c, b^3c\}$	ac	$\{1, c, b^3\}$
ab	$\{b^2c, ab\}$	abc	$\{b^2, abc\}$

Here the row at x is $\{z : xz \in P_{13}\}$. All rows differ, so the bipartite monoid is reduced. Thus the all-multiplicity outcome law proves:

Proposition 6.3. *The exact misère quotient generated by $H_1, \ldots, H_{13}$ is the twelve-element reduced bipartite monoid in the preceding presentation.*

6.3 The second extension at heap 18

The next heaps have values

$$x_{14} = b, \quad x_{15} = a, \quad x_{16} = c, \quad x_{17} = b.$$

Here these assertions are again global, not guesses made inside Q_{13}. With 0 denoting the terminal game, the split rule gives

$$\begin{aligned} \mathrm{opts}(H_{14}) &= \{c, ab, a, 0\}, & \mathrm{opts}(H_{15}) &= \{c, b, 0\}, \\ \mathrm{opts}(H_{16}) &= \{ac, b^2, b, a\}, & \mathrm{opts}(H_{17}) &= \{c, ab, a, 0\}. \end{aligned}$$

The reversible-option reductions of Grundy and Smith give respectively $H_{14} \equiv_- b$, $H_{15} \equiv_- a$, $H_{16} \equiv_- c$, and $H_{17} \equiv_- b$; these are also the four reductions printed in [1, p. 421]. Heap 18 has exact option set $\{0, b, c, bc\}$ and creates a generator $d = H_{18}$. By Lemma 6.2, every position generated by $H_1, \ldots, H_{18}$ is globally equivalent to a count state $a^e b^j c^k d^r$, and every such count state occurs in that universe.

The all-multiplicity recursion extends once more:

$$\begin{aligned} a^e b^j c^k d^r \in P \quad \Longleftrightarrow \quad & \big(r \text{ even and } a^e b^j c^k \text{ satisfies the} \\ & \qquad \text{preceding three-generator law}\big) \\ & \text{or } \big(r \text{ odd},\ e \text{ even},\ j = 0\big). \end{aligned} \tag{17}$$

To check the recursion, use $d \to \{1, b, c, bc\}$. If r is even and the three-generator state is P, the preceding proof handles the a, b, c moves, while every d move has odd r and either odd e or positive j. If the three-generator state is N, use its displayed winning move. The only case with no a, b, c component is terminal when $r = 0$; for positive even r, the move $d \to 1$ reaches the odd-r clause.

If r is odd, a state in the displayed P clause has even e and $j = 0$. An a move makes e odd, a c move makes either e odd or j positive, and a d move lands, at even r, in one of

$$(\epsilon, j, k) = (0, 0, k), (0, 1, k), (0, 0, k+1), (0, 1, k+1),$$

all of which fail the three-generator criterion. Conversely an odd-r N-state has a P move in the following exhaustive cases:

ϵ	j	move	ϵ	j	move
1	0	$a \to 1$	0	1	$b \to 1$
0	even ≥ 2	$d \to 1$ if k even; $d \to c$ if k odd	0	odd ≥ 3	$d \to b$ if k even; $d \to bc$ if k odd
1	1	$b \to a$	1	even ≥ 2	$d \to bc$ if k even; $d \to b$ if k odd
1	odd ≥ 3	$d \to c$ if k even; $d \to 1$ if k odd			

Each target is one of the three even-r cases already proved. This establishes the outcome formula for all $(e, j, k, r) \in \mathbb{N}^4$.

As before, the formula itself identifies the finite presentation. Before the last relation there are 32 forms $a^\epsilon b^j c^\eta d^\rho$, with $\epsilon, \eta, \rho \in \{0, 1\}$ and $0 \leq j \leq 3$, subject to $a^2 = c^2 = d^2 = 1$ and $b^4 = b^2$. The relation $b^2 = ab^3c$ acts freely in the sixteen-element b^2-ideal, producing eight pairs, and leaves the sixteen forms with $j = 0, 1$ fixed. Hence exactly 24 candidate forms remain, and the outcome law descends to them. The quotient is

$$Q_{18} = Q_{13} \times \langle d \mid d^2 = 1 \rangle, \qquad P_{18} = \{a, ac, b^2, d, cd\}. \tag{18}$$

Its 24 normal forms are the twelve above and their products by d. For an explicit reduction certificate, let R be the ordered set of twelve base forms, put $P_0 = \{a, ac, b^2\}$ and $P_1 = \{1, c\}$, and define

$$A_q = \{z \in R : qz \in P_0\}, \qquad B_q = \{z \in R : qz \in P_1\}.$$

The first coordinate A_q is the Q_{13} row already displayed. The second coordinates are

q	B_q	q	B_q	q	B_q
1	$\{1, c\}$	c	$\{1, c\}$	b	$\varnothing$
bc	$\varnothing$	b^2	$\varnothing$	b^2c	$\varnothing$
b^3	$\varnothing$	b^3c	$\varnothing$	a	$\{a, ac\}$
ac	$\{a, ac\}$	ab	$\varnothing$	abc	$\varnothing$

The translation row of q in Q_{18} is (A_q, B_q), while that of qd is (B_q, A_q). The twelve displayed pairs and their twelve swaps are all distinct, so every two normal forms have a separating context. Hence:

Theorem 6.4 (Exact Grundy prefix)**.** *The exact misère quotient generated by $H_1, \ldots, H_{18}$ is the 24-element reduced bipartite monoid in the preceding presentation.*

The statements concern entire submonoids generated by the indicated heaps; no multiplicity bound is present. They do not decide whether the full Grundy quotient is finite. They do show that the apparent period-three tame pattern fails for an exact, explicit reason at heap 13, and they give the first two symbolic extension layers for a future infinite-family or stabilization argument.

Only the individual-heap reductions displayed above are global. The relations $c^2 = d^2 = 1$, $b^4 = b^2$, and $b^2 = ab^3c$ are conclusions inside the localized quotient Q_{18}; they are not asserted as identities in the global semigroup of all misère impartial games. (The relation $a^2 = 1$ is global.)

6.4 A canonical mate missing from the Grundy universe

This obstruction concerns the global values available as Grundy positions, not the finiteness of their localized quotient. It rules out one possible route from global canonical distinctions to an infinite family of Grundy contexts.

We use global impartial misère equality throughout this subsection, and write $G \approx H$ when $G = H$ or $G = H + a$, where $a = *1$. We need the following standard results, all with proofs or precise statements in [10]: the Grundy–Smith simplification rule and simplest form theorem (Definition 2 and Theorems 3, 8, 9), cancellation (Theorem 15), the fact that $0, a$ are the only units (Lemma 17), and the canonical-option consequence of equality (Corollary 21). The latter says that if a canonical game G equals a game Y, then every canonical option of G equals an actual option of Y. Indeed, apply Corollary 21 to $G + 0 = Y$; the alternative requiring an option of 0 is unavailable. We call this the canonical-option lifting property. We also use the equality test (Lemma 11 and Theorem 12): if $H = G$, then every actual option O of H either equals an option of G or has an actual option equal to G.

We will also use the following elementary prime criterion, spelling out its nonunit hypothesis: a *nonunit* canonical game with 0 or a as an option is prime up to the units $0, a$. To see this, write $G = X + Y$ and lift its unit-valued option by canonical-option lifting; one of X, Y is then a part of a unit, hence a unit. This is the argument of [10, Proposition 34]. Including "nonunit" avoids applying the criterion to $a = \{0\}$ itself.

Put

$$a = \{0\}, \qquad b = \{0, a\}, \qquad c = \{a, b, b^2\}, \qquad k = \{0, b, b^2\}.$$

Here $b^2 = b + b$ is a disjunctive sum, not a Nim multiplication. The global reductions already established for Grundy's game give

$$H_{13} = H_{16} = c, \qquad H_{10} = 0,$$

and every heap through 13 has global value in $\{0, a, b, c\}$.

Lemma 6.5. *The games c, k are canonical nonunits, are not associates of b, and $k + a$ is canonical with k as an option. Moreover, k is the mate of the canonical game c.*

Proof. The needed elementary canonical forms are

$$ab = \{0, a, b\}, \qquad b^2 = \{b, ab\}, \qquad ab^2 = \{b, ab, b^2\}.$$

For ab, the raw option set is $\{b, a, 0\}$; for b^2, it is $\{b, ab\}$; and for ab^2, it is $\{b, ab, b^2\}$. The simplification rule verifies that these displayed forms are canonical. Here are the checks explicitly. The possible simplification targets for ab lie in $\{0, a\}$. Both already occur among the options of ab; either would be an extra option required to have itself as an option, which it does not. For b^2, the possible targets are $0, a, b$. The target 0 fails the misère proviso, since b^2 is P; the other targets require the missing option 0. For ab^2, the candidates are $0, a, b, ab$: 0 fails because b^2 has no 0 option, and each nonzero candidate requires the missing 0 option.

For c, every possible proper simplification target lies among $0, a, b, ab$. The target 0 fails because b^2 has no 0 option; each nonzero target requires the missing option 0. Thus c is canonical. For k, the possible targets are likewise $0, a, b, ab$. The target 0 fails because 0 itself is an option. The target a fails because the extra option b^2 has no a option. The targets b, ab require the missing option a. Thus k is canonical.

Adding a to k gives the raw option set

$$k + a = \{k, a, ab, ab^2\}.$$

Its possible proper simplification targets are $0, a, b, ab, b^2$. The target 0 fails because ab^2 has no 0 option. The targets a, b, ab require the missing option 0, while b^2 requires the missing option b. Thus this form is canonical and retains k. The displayed distinct canonical forms also show that c, k are nonunits, that c is not b or ab, and that no element of $\{0, a, b, c\}$ is an associate of k.

For the mate assertion, the definition $0^- = a$ and $G^- = \{(G')^- : G' \in G\}$ gives $a^- = 0$, $b^- = b$, and $(ab)^- = ab$. Consequently $(b^2)^- = \{b, ab\} = b^2$, and $c^- = \{0, b, b^2\} = k$. □

Lemma 6.6. *For every finite impartial game X,*

$$c + X \notin \{0, b, b^2\}.$$

Proof. Both b and c are nonunit primes by their unit-valued canonical options. The equality $c + X = 0$ would make c a unit. The equality $c + X = b$ would, by primality of b, make c a unit or an associate of b. Both possibilities contradict the displayed canonical forms.

Suppose $c + X = b^2$. The canonical option b of b^2 lifts to an actual option of $c + X$. An option of the form $c + X'$ cannot equal b, by the previous paragraph. Therefore $c' + X = b$ for one of the three canonical options $c' \in \{a, b, b^2\}$. If $c' = a$, then $X = ab$, and cancellation in $c + ab = b^2$ yields $c = ab$, a contradiction. If $c' = b$, cancellation gives $X = 0$, whence $c = b^2$, again contradicted by the canonical forms. If $c' = b^2$, cancellation in $b^2 + X = b$ gives $b + X = 0$, contradicting that b is a nonunit. This exhausts all possibilities. □

Theorem 6.7 (A missing canonical mate)**.** *No position of Grundy's game is globally equivalent to k or $k + a$. In particular, the image of the Grundy universe in the global monoid of impartial misère values is not closed under taking the mate of a canonical representative.*

Proof. Suppose otherwise, and choose a Grundy position Y globally associated to k minimizing lexicographically

$$\left(\sum_{h\in Y} h, \quad \sum_{h\in Y} \max(h-2,0)\right).$$

A legal split preserves the first coordinate and strictly decreases the second. Deleting any positive-size heap strictly decreases the first. The game k is a nonunit prime because its canonical form has 0 as an option. The same is true up to association for $k + a$. Hence in any additive decomposition of Y into heaps, all but one component are units, and the remaining heap is associated to k. Minimality therefore forces Y to be a single heap H_n.

If $H_n = k + a$, its canonical option k lifts by canonical-option lifting to an actual option of H_n, contradicting minimality. Thus $H_n = k$. The equality test now says that every actual option of H_n either has value in $\{0, b, b^2\}$ or has an actual option of value k. The second case gives a two-move follower of H_n of smaller rank and contradicts minimality. Thus every legal split of H_n has global value in $\{0, b, b^2\}$.

For $n \geq 14$ with $n \neq 26$, the unequal split $13 + (n - 13)$ is legal and has value $c + H_{n-13}$, contradicting Lemma 6.6. For $n = 26$, the split $10 + 16$ has value $0 + c = c$, giving the same contradiction. Finally, the known global reductions for $n \leq 13$ have values in $\{0, a, b, c\}$, none associated to k. All heap sizes have been excluded. □

Remark 6.8. The mate operation is not invariant under changing a game to a globally equivalent form; see [10, Section 3]. Thus the theorem is deliberately about the mate of the *canonical representative* c, not a claim that the literal unreduced tree H_{13}^- has value k. Nor does the theorem exclude localized testers or prove the full Grundy quotient finite or infinite.

6.5 A square-option constraint on finite Grundy traces

The split rule and finite coloring locate a particular heap transition with a nonconstant power cycle. This is a constraint on the heap trace; generic nontriviality of a finite quotient's kernel already follows from Proposition 2.4 and supplies no additional Grundy obstruction.

Theorem 6.9 (A square-option heap in every finite Grundy trace)**.** *Suppose the full misère quotient Q of Grundy's game is finite. Then there is a heap transition (c, E) with $c^2 \in E$. Every two consecutive positive powers of c are distinct, and the powers of c are eventually periodic with a period of length at least two.*

Proof. Color each positive integer n by the heap value $x_n \in Q$. A standard distinct-summand form of Schur's theorem gives $i < j$ for which

$$x_i = x_j = x_{i+j} = c.$$

For completeness, it follows directly from finite Ramsey: color the edge $\{r, s\}$ by $x_{|r-s|}$ and take a monochromatic K_4 with vertices $r_0 < r_1 < r_2 < r_3$. Splitting the outer difference at r_1 and at r_2 gives two monochromatic Schur triples; their two summands cannot both be equal, since that would force $r_1 = r_2$. Choose the unequal one.

Because $i < j$, the position $H_i + H_j$ is a legal option of H_{i+j}. Consequently $c^2 \in E$. Closure of the transition table puts the r-fold transition power

$$(c^r, c^{r-1}E)$$

in the table for every $r \geq 1$. Its option set contains c^{r+1}. Proposition 2.3 therefore gives $c^r \neq c^{r+1}$ for every $r \geq 1$. Finiteness of Q makes the power sequence eventually periodic, and the displayed inequalities exclude period one.

□

The conclusion is structural rather than contradictory: period two already occurs in tame quotients. To prove the full Grundy quotient infinite, one still needs either unbounded kernel rank or infinitely many distinct translation rows, not merely one nonidentity kernel element.

7 Authority and exact frontier

The abstract classification theorem is Siegel's cited result. The new claims in this paper have the authority recorded in Table 1. The distinctions are substantive: a bounded quotient signature is not an exact infinite-ruleset certificate, and a Lean definition of a transition target is not a proof that a natural ruleset realizes it.

Status	Content
Cited input	Plambeck–Siegel quotient theory, Siegel's valid-transition-table classification, Ferguson's subtraction theorem, the stated Grundy–Smith global reductions, global misère canonical forms and cancellation, and the unary conjunctive-grammar hardness theorem.
Proved here	Option-set determinism; the exact trace criterion and digit exclusions; no-split periodicity, periodic split certificates, and periodic witness compression; arbitrary- decoder hardness and valid-table parity contrast; two-element split rigidity and mask classification; the finite-exception normal form and inert-pad obstruction; complete subtraction synthesis; the exact Grundy quotients through heap 18; the missing canonical mate; and the square-option theorem.
Kernel-checked	Transition determinism, meximality, P-reaching contexts and no-self; the star involution and free action; exact quotient fibers; octal record periodicity and tail certification; safe saturation, terminal prefix repair, and the actual compressed masks with all-heap validity; odd-removal totality; the binary split period and finite-candidate criterion; the actual subtraction mex sequence and all-position outcome law; the typed prefix/pad/unary-tail construction; the inert-pad obstruction; the unary-language dictionary; and the finite-color and power-period cores of the Grundy constraint.
Paper-level bridge	The grammar reduction itself; numerical enumeration and multiset transport for the typed normal form; the explicit two-element mask classification; the exact-quotient interpretation of compressed records and the bounded synthesis algorithm; the subtraction quotient presentation, reduction, and synthesis algorithm; and the Grundy presentations, translation rows, and canonical-mate obstruction.
Open	Fixed valid-table split totality, unbounded finite-code synthesis, the tame extensions of $\mathcal{R}_8$, and the exact full quotient of misère Grundy's game.

Table 1: Claim authority for the natural-realization paper.

The import-only entry point `MisereNaturalRealization` contains no parallel paper-specific copy of the transition calculus. The theorem owners are separated by role:

- `MisereTransition`: transition algebra and least-rank determinism;
- `MisereOctalCertificate`: finite octal certificates;
- `MisereNaturalUniversality`: typed realization and normal forms;
- `MisereTraceLanguage`: the partial-trace reduction;
- `MiserParityTrace`: universal star structure, odd-removal totality, and the finite binary split criterion;
- `MisereSaturation`: parity-preserving option-set enlargement;
- `MisereOctalTerminal`: the terminal-prefix repair;
- `MisereOctalCompression`: actual compressed masks, their length bound, and all-heap record validity;
- `MisereSubtraction`: the mex sequence, Ferguson shift, and all-position misère outcome theorem; and
- `MisereGrundyObstruction`: the finite-color square-option and power-period constraints.

The entry point imports only the terminal owners needed to expose this dependency graph.

The remaining problem has a fixed-trace decision layer and a synthesis layer. For fixed finite (Q, P), valid table T, and split code Γ, one must decide whether the canonical partial trace is total. The arbitrary-decoder Π^0_1 theorem shows why a proof must use validity, but gives no hardness result for valid tables; indeed its own code is total in every nontrivial valid table. Two-element split targets are completely decided by Theorem 3.11.

For synthesis, Theorem 3.6 removes an independent code-length gap within the periodic branch: bounding the preperiod and period already bounds a realizing code. Table enumeration adds no further gap, because only finitely many transition tables exist on fixed Q. What remains is a computable bound on the necessary periodic complexity, and an argument accounting for realizations whose heap words might not be ultimately periodic. An undecidability proof for synthesis would be an alternative. Ordinary subtraction codes already admit complete synthesis by Theorem 5.3, but this subclass is not asserted to be maximal among decidable natural classes.

For $|P| = 2$, the tame arm is complete and the tame extensions of $\mathcal{R}_8$ remain. For Grundy's game, a complete solution must either continue the exact presentations and prove stabilization, or exhibit infinitely many distinct translation rows with explicit separating contexts. In the finite branch, Theorem 6.9 requires a square-option heap and a nonconstant eventual power cycle. The universal star involution prevents using generic kernel nontriviality as a further obstruction. Theorem 6.7 excludes transferring global distinctions by assuming closure under canonical mates; any separating contexts must instead be constructed inside the Grundy universe.

8 Conclusion

Natural realization is a totality and synthesis problem for one canonical quotient-valued heap trace. No-split codes are finite-state; split codes add the unbounded convolution measured by the unary-language equations. The finite-exception normal form shows that an infinite uniform tail is not by

itself the obstruction. Strict octal homogeneity is: one digit couples an intended prefix move to infinitely many translated moves.

That formulation supports exact positive and negative results. The no-self and inert-pad theorems reject impossible direct encodings, while ordinary subtraction admits a complete quotient and synthesis classification. Periodic witness compression also gives finite synthesis for bounded preperiod and period. Validity forces universal parity on odd-removal codes and period two on every two-element split trace. Grundy's game supplies exact wild prefixes at heaps 13 and 18, but no stabilization or infinite separation theorem yet follows. The next decisive result must control the valid split recurrence or the Grundy extension process uniformly; further bounded tables alone cannot close either problem.

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